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CARDIOLINK: A GEMINI-BASED PPG-ECG INTEGRATED PHYSIOLOGICAL MONITORING SYSTEM

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ABSTRACT OF THE STUDY

Healthcare accessibility remains a critical challenge, especially in underserved and remote regions where a shortage of medical professionals and the high cost of routine check-ups delay critical interventions. This gap in continuous health monitoring necessitates affordable, real-time solutions. To address this, we developed CardioLink, a wearable health monitoring system. It integrates an ESP32-S2 microcontroller with sensors that capture vital signs: a MAX30102 pulse oximeter uses light-based photoplethysmography (PPG) for heart rate and SpO_2; an MLX90614 infrared sensor detects skin temperature without contact; and an AD8232 module acquires electrocardiogram (ECG) signals via electrodes. The system operates on an Input-Process-Output model: it collects raw sensor data, the microcontroller processes and filters it into readable metrics, and then transmits the information securely to a cloud database for remote access by healthcare providers. Experimental trials demonstrated that CardioLink achieves an average sensor accuracy of 98.7% with a rapid 2-3 second response time. By providing a reliable and timely tool for early abnormality detection, CardioLink offers a viable solution to enhance proactive patient care and reduce the burden on congested healthcare systems.

Keywords: Healthcare Accessibility, Wearable Health Monitoring, CardioLink System, ESP32, Real-time Monitoring

Chapter I

Context and Rationale

Background of the Study

Access to healthcare remains a significant challenge, largely due to the lack of immediate health monitoring systems. It is estimated that over 4.5 billion people, or more than half of the global population, lack full access to essential health services (World Health Organization, 2021). This includes critical areas such as preventive care and disease management, both of which are vital for one's well-being. As a result, many individuals are forced to postpone medical responses due to the lack of timely health monitoring systems, which can lead to undiagnosed conditions and delayed treatments. Additionally, the exorbitant cost of healthcare makes frequent checkups difficult to afford, causing individuals to delay early intervention and, in turn, experience more severe complications.

Developed nations have made significant advancements in incorporating smart medical monitoring technologies into their healthcare systems, thus facilitating early disease diagnosis and patient care management. However, most low and middle-income countries, including most ASEAN nations, still experience challenges concerning accessibility and affordability. Apart from that, healthcare staff shortages worsen the scenario, as some ASEAN nations indicate that fewer than 0.1% of the population work as licensed health professionals, leading to delayed patient care and reduced medical services efficiency (Alibudbud, 2023; Corpuz, 2023).

In the Philippines, approximately 60% of regions lack adequate health services, especially in rural and remote areas (Royo et al., 2023). Access to

continuous and accessible health monitoring remains a significant challenge in the Philippines, especially for individuals with chronic illnesses, the elderly, and those living in rural or underserved communities (Abalos, 2018). Healthcare accessibility remains a significant challenge; an estimated 60% of regions lack basic health services, further exacerbating the issue (Domingo et al., 2021). Factors such as rural-urban disparities, limited funding, and a shortage of healthcare workers contribute to these accessibility barriers (Dondonayos et al., 2023). Although telemedicine programs have been implemented by public and private institutions, the cost of routine medical checkups also acts as a deterring factor, leading to unnecessary health complications that further congest the healthcare system in the country. Addressing this gap can enable patients to receive timely and accurate diagnoses, leading to early intervention and better management of chronic conditions. This, in turn, can help reduce complications, hospitalizations, and even save lives. Additionally, real-time monitoring systems would allow healthcare providers to allocate resources more efficiently, prioritizing those in urgent need while preventing healthcare system overload.

To address the stated limitations, CardioLink, a health monitoring system based on ESP32, has been proposed as an innovative solution. This system is designed to enhance patient care by providing continuous and real-time monitoring of vital signs. Equipped with sensors, CardioLink tracks health parameters such as body temperature, heart rate, oxygen levels (SpO_2), and electrocardiogram (ECG), ensuring timely detection of abnormalities. The device is ergonomically designed as a wearable armband, making it comfortable for all-day use and ensuring consistent sensor contact for accurate readings. The collected data is securely transmitted to registered medical

practitioners through a PHP-based cloud database, allowing for remote monitoring and early intervention. CardioLink features an in-built display on the armband that provides instant readings, allowing users to stay informed about their health status at a glance. With the integration of Artificial Intelligence, using Gemini 2.0 Flash, it can analyze potential health risks and suggest precautionary measures. Additionally, the device includes an alert mechanism that immediately notifies family members and medical professionals in case of any detected abnormalities, enabling a swift medical response. By integrating modern technology with convenient, armband-based real-time health tracking, CardioLink aims to revolutionize remote patient monitoring, enhance healthcare accessibility, and ultimately contribute to improved disease management and overall well-being.

Conceptual Framework

This conceptual framework illustrates how CardioLink's real-time health monitoring system was assessed, outlining the relationship between key variables to ensure a well-structured evaluation. The use of the Input-Process-Output (IPO) Model elaborates how data is collected (input), analyzed (process), and converted into meaningful insights (output) to ensure CardioLink's accuracy and effectiveness.

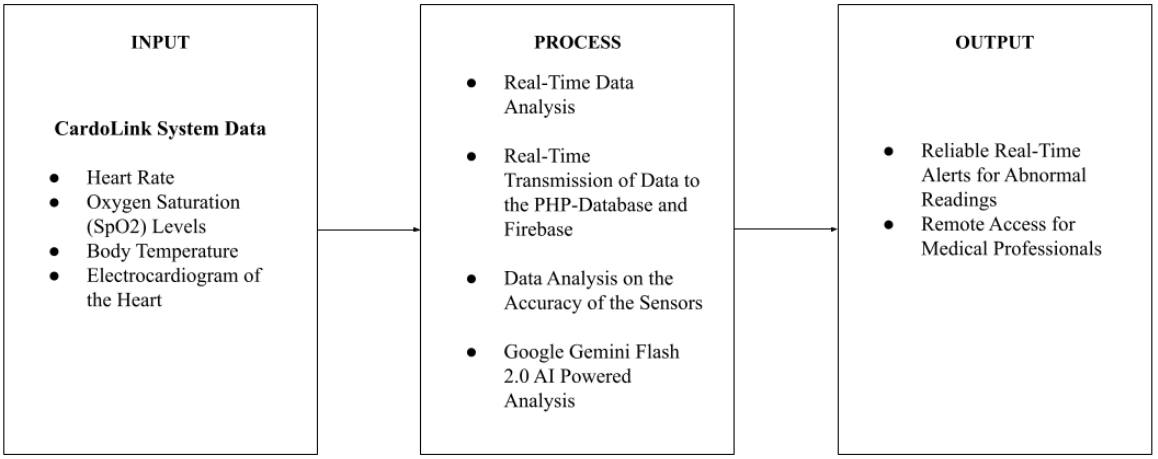

Figure 1. Research Paradigm

The Input-Process-Output (IPO) Model is used to explain how the device collects, manages, and utilizes health data. Through sensors integrated into the armband, CardioLink gathers health information, ensures its accuracy, and continuously monitors the patient's vital signs in real time. Heart rate and oxygen saturation are measured using the MAX30102 pulse oximeter sensors within the armband, which analyze blood flow and determine oxygen levels in the blood. The system also measures body temperature without direct contact using the MLX90614 infrared temperature sensor, also housed in the armband. In addition, the system measures electrocardiogram readings using the AD8232 ECG Measurement Module, with electrodes designed for seamless contact via the armband. The ongoing monitoring from these sensors helps detect potential health issues early.

The ESP32-S2 microcontroller, which is the heart of the system and embedded within the armband, analyzes and sends the data collected during the process step. To confirm the readings and check if any of the patient's vital signs and electrophysiological signs are abnormal, real-time data analysis is performed. After

the measurements are taken, the data is sent to a PHP-based cloud database for storage. This data transmission enables medical professionals and caregivers to remotely monitor the patient, ensuring the information is always available when needed.

Additionally, the system checks the accuracy of the sensors by comparing the current readings with the reference values, ensuring the armband device is reliable for monitoring health. The output process gathers and processes the data to provide users and healthcare professionals with accurate, real-time alerts for abnormal vital signs and electrocardiogram readings. If there are any issues with the body temperature, heart rate, SpO_2, or electrocardiogram readings, the armband device immediately notifies the user and medical staff. Furthermore, by providing access to the patient's records through the PHP-based database, doctors and caregivers can reduce the need for in-person hospital visits. CardioLink serves as a valuable tool for real-time health tracking from a convenient armband, enhancing patient safety and encouraging early diagnosis, ultimately reducing the need for hospital admissions.

System of the Study

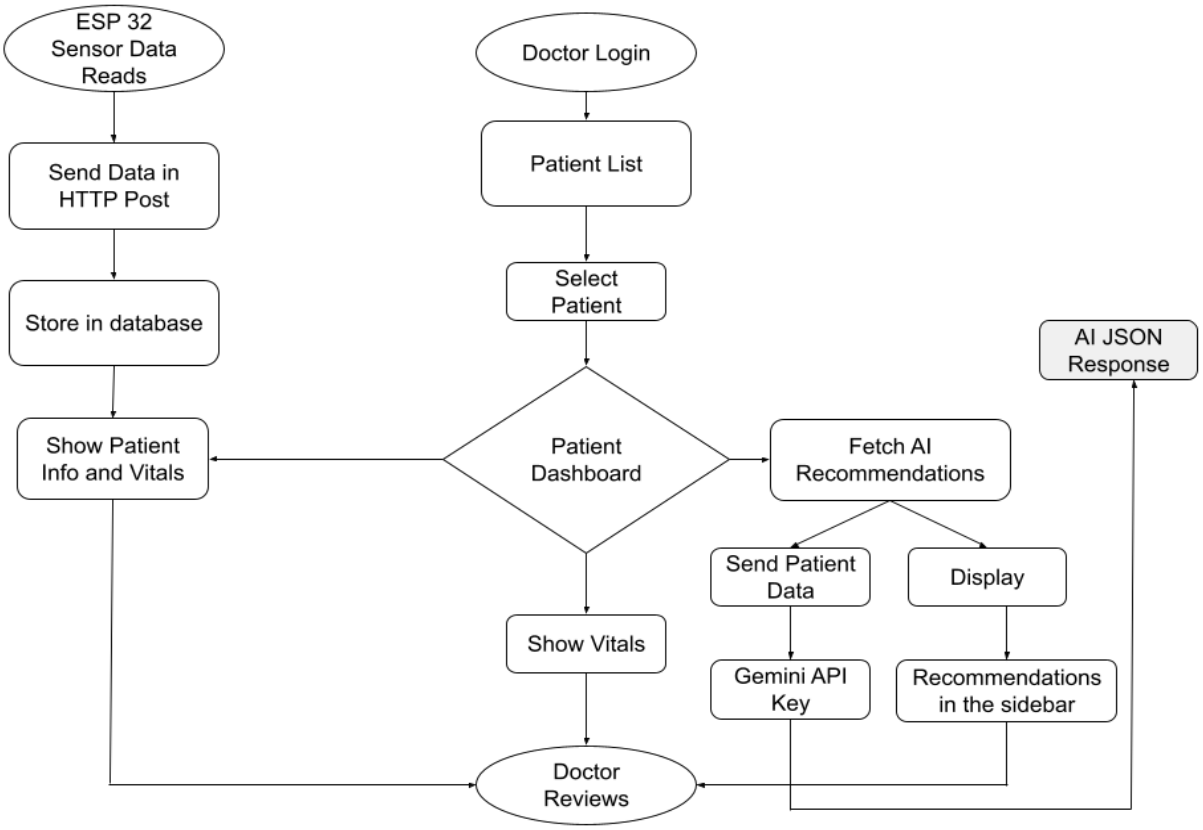

Figure 2. System of the Study

CardioLink is a wearable health tracker designed to keep an eye on vital signs in real time. It measures heart rate, body temperature, oxygen levels, and the electrocardiogram of the heart using built-in sensors, and then sends the data to a small but powerful ESP32-S2 microcontroller. From there, the information is stored securely in a PHP-based database, making it easy to access when needed.

Users can check their health stats right on the device's screen, while caregivers and family members can monitor updates remotely through an app. If everything looks normal,the system quietly continues tracking. But if it detects something

unusual, like a dangerously high temperature or low oxygen levels, it immediately sends an alert to notify the user and their loved ones, allowing for quick action.

By automating health monitoring and providing instant alerts, CardioLink helps take the guesswork out of tracking vital and electrophysiological signs. Whether people with medical conditions or anyone who wants peace of mind, the system makes remote health monitoring easier, and more reliable.

Statement of the Problem

The study aims to develop CardioLink, a device that provides real-time health monitoring systems such as heart rate, oxygen levels (SpO_2), body temperature, and electrocardiogram (ECG) readings. Specifically, it will seek to answer the following:

1. What is the accuracy of CardioLink's measurements in terms of:
   1.1 Heart Rate (BPM);
   1.2 Oxygen Saturation (SpO_2);
   1.3 Body temperature (°C); and
   1.4 Electrocardiogram (ECG) QRS Complex (ms)?
2. What is the response time of CardioLink in sending real-time data to the Firebase in terms of:
   2.1 sensor data to the on-board display; and
   2.2 Transmission of sensor data to the Firebase?
3. How do the criterion participants evaluate the sensor accuracy and response time of CardioLink?
4. Is there a significant difference between the measurement of the already existing physiological measuring devices and CardioLink?

5. Implications of the study for the development and improvement of real-time health monitoring systems.

Research Hypothesis

With technology improving healthcare, it's crucial to see if CardioLink is as accurate as traditional physiological monitoring medical devices, but improved than existing physiological measuring devices. This study aims to compare its measurements of heart rate, oxygen saturation (SpO_2), body temperature, and electrocardiogram (ECG) readings, leading to the following hypothesis:

Null Hypothesis (Ho): There is no significant difference between the measurements obtained from CardioLink and those from existing physiological measuring devices in terms of heart rate, oxygen saturation (SpO_2), body temperature, and electrocardiogram (ECG) readings.

Scope and Delimitation

This study focuses on the development and design of CardioLink, a wearable armband health monitoring system that is designed for real-time monitoring of heart rate, blood oxygen levels (SpO_2), electrocardiogram (ECG) readings, and body temperature. The system completely integrates an ESP32-S2 microcontroller with MAX30102 (pulse oximeter), MLX90614 (temperature sensor), and AD8234 (ECG Sensor) to collect all the real-time health data from the sensors. The collected data is transmitted into a PHP-database, and transmitted through a cloud that healthcare professionals are authorized to monitor their vital signs. Additionally, this sends a notification if there is an abnormal vital sign that alerts healthcare providers and

caregivers. The study also includes the development of the prototype of CardioLink. Additionally, the system is integrated with an artificial intelligence powered by Gemini 2.0 Flash for health risk detection and sends it to healthcare providers and caregivers in case of abnormal vital readings. However, there are delimitations in this study. It is only limited to monitoring heart rate, SpO2, body temperature, and electrocardiogram (ECG) readings, and does not include blood pressure. The ECG readings only focus on the QRS readings since it is the most clinically significant indicator of cardiac activity. Additionally, the system is strictly precautionary and does not describe medication or perform medical diagnosis or predictive treatment. Consulting a healthcare provider is still needed. Testing was conducted in a controlled environment with stable internet connections, which is highly considered; real-time and poor connectivity with environmental factors are less considered.

Significance of the Study

This study had mostly benefit healthcare professionals and caregivers, as well as patients. Additionally, it improves healthcare accessibility. This study provides an efficient solution for health monitoring.

Caregivers. By providing doctors and nurses with continuous health insights, the system improves early detection, lowers hospital readmissions, and optimizes treatment plans, ultimately improving patient care.

Health Care Professionals. It gains access to patient monitoring and also allow them to assess their patients' health conditions more efficiently. This is crucial in rural areas where healthcare access is limited and shortage of medical personnel.

Medical Institutions. Integrating CardioLink into healthcare operations allows institutions to improve workflow efficiency, allocate resources more effectively, and improve patient outcomes, particularly in underserved areas with limited healthcare infrastructure.

Future Research. Studies could look into the device's development in reducing emergency hospital visits and improving overall health outcomes for individuals. These insights could help to shape the future of telemedicine and personalized healthcare solutions.

Definition of Terms

To ensure a better understanding of the study, the following terms are defined technically and operationally:

Body Temperature Sensor. A sensor that tracks body temperature to monitor fever, infections, or other health conditions.

CardioLink. A health monitoring device powered by the ESP32-S2 microcontroller, designed for real-time tracking of vital signs and secure data transmission to healthcare providers and caregivers.

ECG Measurement Module. A sensor tracks the electrophysiological sign of the heart that monitors potential heart muscle strain or heart attack.

ESP32-S2 Microcontroller. A low-power, Wi-Fi-enabled microcontroller used in CardioLink to facilitate wireless data transmission and processing.

Healthcare Accessibility. The ease with which individuals, particularly those in low-income and rural areas, can obtain essential medical services and health-monitoring tools.

Health Monitoring Device. A wearable armband electronic device equipped with sensors to track physiological signs such as body temperature and blood oxygen levels in real-time.

PHP-Based Database. A cloud-based database that stores and manages real-time data in CardioLink.

Pulse Oximetry Sensor. A sensor that measures oxygen saturation (SpO2) in the blood, helping detect conditions such as hypoxia and respiratory issues.

QRS Complex. A waveform on an electrocardiogram (ECG) that represents the depolarization of the heart's ventricles, causing them to contract.

Vital Signs. Key physiological measurements that indicate an individual's health status, including body temperature, pulse rate, respiratory rate, and blood oxygen levels.

Chapter II

Materials and Methods

Study Objects or Materials

Table 1.

Description and Function of Materials or Objects

Component/ Material	Image	Description	Functionalities
ESP32-S2 Microcontroller		Low-power, Wi-Fi-enabled microcontroller	Facilitates wireless data transmission and processing
MAX30102 Pulse Oximeter Sensor		An optical sensor that measures heart rate and oxygen saturation (SpO_2)	Monitors pulse rate and SpO_2 levels
MLX90614 Infrared Temperature Sensor		Contactless infrared sensor for body temperature measurement	Measures body temperature without skin contact
AD8232 ECG Sensor		Measure the electrical activity of the heart to produce an electrocardiogram (ECG).	Amplifying, filtering, and extracting the heart's weak electrical signals, or biopotentials, to produce a usable ECG waveform

2.8-inch Display		Small LCD screen with a graphical interface	Displays real-time sensor data (heart rate, SpO_2, temperature)
Plastic PVC Casing		Lightweight, durable housing for electronic components	Protects internal components and ensures device durability
Wearable Strap		Adjustable strap for securing the device on the user's arm	Allows comfortable and secure placement for daily use

The table above represents the components and materials that are being used in the building process of CardioLink's health monitoring system, alongside their description and functionalities. Each of the components plays a vital role in monitoring the device's accuracy, efficiency, and reliability in tracking the vital signs, specifically the heart rate, oxygen saturation, body temperature, and electrocardiogram of the heart (ECG).

This study focuses on the CardioLink health monitoring device, which is a wearable armband system designed to track vital health parameters in real time. The device is based on the ESP32-S2 microcontroller, which allows wireless data transmission.

This connectivity ensures that authorized healthcare providers and caregivers have secure access to real-time health data. The primary sensors integrated into the device include the MAX30102 pulse oximeter, which measures heart rate and blood oxygen levels (SpO2), and the MLX90614 infrared temperature sensor, which monitors body temperature. These sensors are crucial in assessing a user's overall health status and detecting abnormalities that may require medical attention.

The device's physical structure is designed to be both wearable and durable. It is enclosed in a plastic-PVC housing that is both lightweight and durable, protecting the internal electronic components from environmental damage.

The wearable strap, like a traditional pulse oximeter, allows users to comfortably secure the device on their arm, making it suitable for daily use. The integration of advanced microcontrollers, accurate medical sensors, and a secure database system enhances its capability as a health monitoring tool, particularly for individuals who require continuous health tracking, such as patients with cardiovascular or respiratory conditions.

Preparation of Object or Materials

Before beginning to build the CardioLink design, the study conducted actual planning on the system's design and functionality. After the plan was developed, the researchers consulted with a robotics specialist. Following receipt of the recommendations, the researchers finalized the system's design and functionality. Once the design had been finalized, the researchers gathered the necessary materials, which included the system and casing. It began with the hardware components, starting with the ESP32-S2 microcontroller, where the entire program was stored for

monitoring. The sensor monitors the heart rate, body temperature, SpO2, and electrocardiogram of the heart (ECG). The next step was to connect it to the 2.8-inch display, which was used to display system visuals.

The next step is to integrate the PHP database, which stores all of the data and displays a graphical interface for vital signs and status. The next step was to fabricate the plastic PVC casing, which was finished and connected with a wearable strap for use. Following the development of CardioLink, the system's sensors were calibrated and tested to ensure accuracy and efficiency, making the device reliable for real-time health monitoring.

Research Design

This study adopted a pre-experimental one-shot case design approach because it provided a controlled setting for a focused assessment of the CardioLink system, allowing vital sign measures to be directly compared with traditional physiological medical measurements. Because it allows for performance evaluation without the complications of a control group. A quantitative pre-experimental one-shot case design was used in this study to evaluate the quality of CardioLink's monitoring of vital signs, including body temperature, heart rate, oxygen saturation (SpO2), and electrocardiogram of the heart (ECG). Assessing the system's correctness, responsiveness, and general dependability was the main goal. In order to ascertain the accuracy and reliability of CardioLink's operation, the sensor data were compared with the already existing medical measurements.

Using a criterion sampling technique, the system's ability to transmit and display health data quickly was evaluated and tested to assess its effectiveness in real-time monitoring.

The system's response time was also evaluated in the study since it was essential to figure out how reliable it was in medical emergencies. A survey on a 5-point Likert scale was used to collect input from chosen participants to gauge the device's usefulness and efficacy.

CardioLink Prototype

CardioLink seeks to simplify and improve health monitoring by making it automatic to track vital signs such as heart rate, oxygen levels, body temperature, and the electrocardiogram of the heart (ECG). Conventional processes tend to be difficult, with frequent checks and personal consultations that can be inconvenient, particularly in a medical emergency. CardioLink corrects this issue by monitoring the user's health continuously and feeding information to health providers automatically, minimizing the need for constant monitoring.

This research analyzed significant features such as accuracy, reliability, and usability to test whether the device is effective and pragmatic.

The value of CardioLink is further enhanced by its capacity to function in emergency conditions and rural locations where access to medical care may be limited. For instance, following a natural disaster, hospitals may become congested with patients, which can make it difficult for individuals to receive immediate care. A device such as CardioLink can assist in monitoring a patient's health in real-time, allowing medical professionals and caregivers to keep tabs on their health even from

afar. Its wearable nature, lightweight, and low power usage ensure that it is convenient enough to utilize anywhere, whether in hospitals, homes, or disaster relief camps. With greater accessibility and reliability in health monitoring, CardioLink can enhance the quality of care for patients and save lives.

Research Procedure Steps

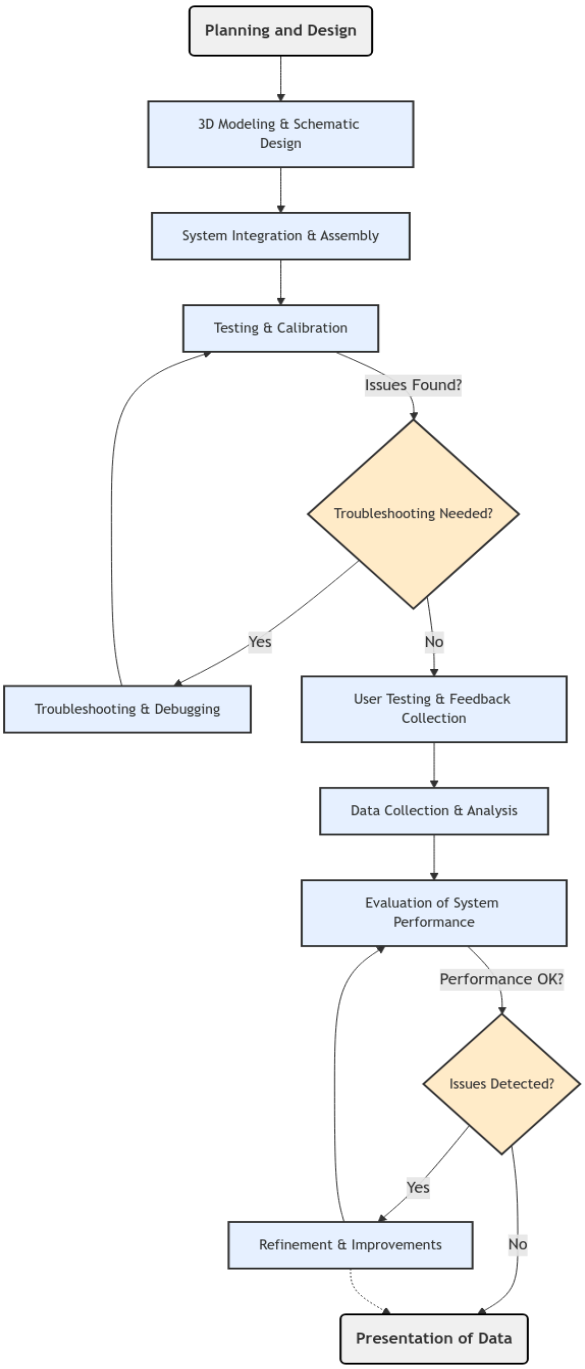

Figure 3. Research Procedure Steps

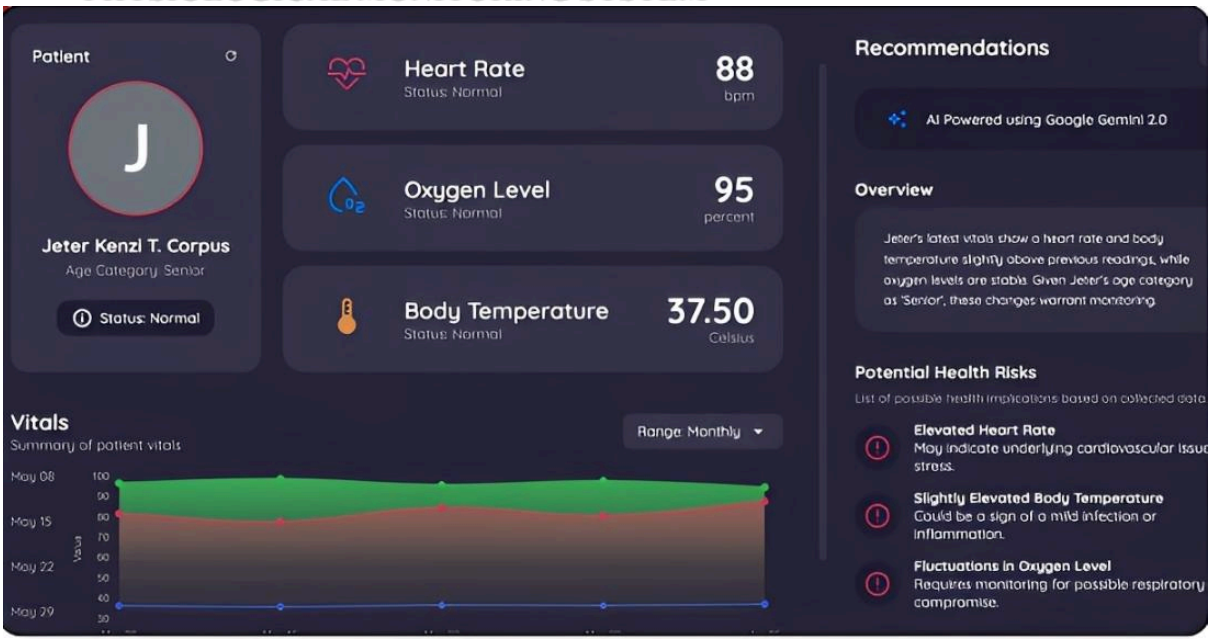

Figure 4. Application Design of CardioLink

After designing the system, the concept was translated into a 3D model and schematic diagram, providing a clear blueprint for integration and assembly. This enabled the researchers to gather the materials and components needed for the system assembly.

Figure 5. 3D Model of CardioLink with Labelled Functionalities

Once the system was put together, it moved into the testing and calibration phase. If any issues were detected, a troubleshooting process was carried out, followed by retesting to ensure proper functionality. If no troubleshooting was needed, user testing and feedback collection were initiated. This step allowed us to gather valuable insights and analyze data to assess the system's performance. If the results did not meet the intended standards, further refinements and improvements were made before collecting additional data. Finally, after gathering all the necessary data, the findings were organized and presented using tables for clearer interpretation.

Ethical Considerations

To ensure respect for ethical standards in this research, comprehensive precautions are implemented, particularly in compliance with the Data Protection Act of 2012. Additionally, this robot, like medical practitioners, is ethically bound to maintain the confidentiality of any information learned during its treatment of a patient. The Code of Ethics of the Philippine Medical Association explicitly states that physicians must treat all patient information as sacred and confidential, even after the patient's death, unless required by law for justice, safety, or public health. Strict measures are taken to protect the confidentiality and privacy of any sensitive data obtained from CardioLink's performance. Adhering to legal requirements, all data collected is securely stored and accessible to authorized personnel only for the sole purpose of investigating this research study. Finally, all data are securely deleted to protect them from unauthorized access, ensuring transparency and ethical stewardship throughout the study.

Methods of Measurement

This study applied quantitative methods to evaluate the effectiveness of CardioLink in real-time health monitoring. It focused on technical accuracy and reliability. These methods are important in assessing the device's ability to monitor heart rate, oxygen saturation (SpO_2), body temperature, and the electrocardiogram of the heart (ECG) with precision. Key metrics include measurement accuracy, data transmission speed, and system responsiveness. The use of quantitative techniques ensures objective comparisons across varying conditions.

CardioLink's monitoring system integrates multiple sensors to detect vital signs and transmit the data to a PHP database. The heart rate monitor, pulse oximeter, and temperature sensor work with an ESP32-S2 to collect readings. Once recorded, the data is sent to a secure database and relayed to a mobile application for real-time access by health authorities.

The accuracy and reliability of these sensors were assessed through comparisons with standard medical devices. Data transmission efficiency was also evaluated by analyzing the speed at which data is processed and displayed. While user-related factors may impact performance, adjustments to the system can enhance its reliability.

To maintain data integrity, the study implemented continuous monitoring and validation procedures. All collected data was securely stored, and any discrepancies were analyzed to ensure consistency. These measures contributed to verifying CardioLink's efficiency and accuracy in real-world applications.

Statistical Analysis

This study used a quantitative approach to evaluate how accurately and efficiently CardioLink functions as a real-time health monitoring device. We focused on measuring how precisely it records heart rate, SpO_2, body temperature, and electrocardiogram of the heart (ECG), as well as how quickly and reliably it transmits data to a PHP-based database and mobile application.

To assess CardioLink's accuracy, the researchers calculated the average values (mean), median, and variability (standard deviation) of its measurements and compared them to those from a clinically validated reference device. The researchers also used percentage proximity to see how close CardioLink's readings were to the standard values.

For real-time data transmission, we measured the average response time and its variability to understand how consistently the device sends sensor data. Additionally, we gathered user feedback through a 5-point Likert scale survey, allowing healthcare professionals and users to share their experiences with the device's accuracy and efficiency.

To also measure the significant difference between CardioLink and traditional physiological monitoring devices. A paired t-test was used to compare the readings from CardioLink and the traditional monitoring devices. It determined if the mean differences between the two measurement sets were significant, hence determining the accuracy and dependability of CardioLink.

Participants of the Study

Purposive criterion sampling was used in this study to choose participants based on attributes that are pertinent to CardioLink evaluation. Five IT specialists, five engineers, and five medical professionals were among the fifteen participants selected. A well-rounded evaluation is ensured by this combination of users who directly benefited from the system and experts in technology, engineering, and healthcare. 52 Tapinac Senior High School teachers age ranges from 22 to 60, also participated in the device's testing.

This method enables in-depth input on CardioLink's precision, responsiveness, and dependability, which aids in improving its functionality and guaranteeing that it satisfies user requirements and industry standards.

Chapter III

Results and Discussions

This chapter covers the findings and discussion of CardioLink's performance based on the survey and trial data collected. The testing focuses on five main aspects: sensory accuracy, response speed, data dependability, system efficiency, and user experience. The accuracy of the sensors that detect the heart rate, oxygen saturation (SpO2), body temperature, and electrocardiogram of the heart (ECG) data. It also measured the transmission speed on the database, which was used during the trials.

1. CardioLink's Sensor Accuracy

The accuracy of CardioLink's sensor was evaluated by comparing the recorded measurements with the reference values. The reference values are the traditional devices used, such as Pulse oximeters, thermometers, and electrocardiograms (ECG), which were provided. The analysis focuses on the four sensors that CardioLink measures, which are the heart rate, oxygen saturation, body temperature, and electrocardiogram of the heart (ECG). Each of them conducted 10 trials to ensure the consistency and reliability of the readings.

From the tables that were given below, this is where the values were presented, including the reference values, sensor measurements, proximity values, and statistical measurements such as the mean, median, and standard deviations. These results provide the accuracy of CardioLink's sensors and reliability in real-time health monitoring

Table 2.

Accuracy of Heart Rate Sensor: Reference and Sensor Data Mean

Trial	Reference Data Measurement	Sensor Data Measurement	Difference
1	75.10	74.12	-1.30%
2	75.07	74.09	-1.31%
3	74.99	74.03	-1.28%
4	75.03	74.07	-1.28%
5	74.99	74.02	-1.29%
6	75.07	74.09	-1.31%
7	74.97	74.00	-1.29%
8	75.00	74.02	-1.31%
9	75.01	74.02	-1.32%
10	75.13	74.13	-1.33%
Total Mean:	75.04	74.06	-1.302%
Median:	75.02	74.05	
Standard Deviation:	0.05358275	0.046773687	

This table presents the performance of the heart rate sensor over 10 trials. The reference measurements average 75.04 BPM, while the sensor records an average of 74.06 BPM. The median values—75.02 BPM for the reference and 74.05 BPM for the sensor—closely correspond to the means, indicating an equal data distribution with minimal impact from outliers. The sensor exhibits a consistent underestimation of approximately 1.30% relative to the reference. The low standard deviations (equal to or less than 0.05 respectively) further support the reliability and precision of the measurements.

Additionally, our system achieves an overall accuracy of 98.7%, which is a notable improvement compared to the 95.8% accuracy reported by Mustiko and Sri (2023) using a Gaussian Naive Bayes approach. This improvement can be attributed to enhanced calibration and real-time data processing using the MAX30102 sensor.

Table 3.

Accuracy of Oxygen Saturation Level Sensor: Reference and Sensor Data Mean

Trial	Reference Data Measurement	Actual Data Measurement	Difference
1	98.61	97.42	-1.21%
2	98.66	97.48	-1.20%
3	98.58	97.40	-1.20%
4	98.69	97.50	-1.21%
5	98.53	97.36	-1.19%
6	98.61	97.43	-1.20%
7	98.54	97.36	-1.20%
8	98.71	97.52	-1.21%
9	98.57	97.40	-1.19%
10	98.64	97.46	-1.20%
Total Mean:	98.61	97.43	-1.201%
Median:	98.61	97.425	
Standard Deviation:	0.060955357	0.055787294	

In the accuracy of the oxygen saturation sensor, 10 trials indicate a reference mean of 98.61% and a sensor mean of 97.43%. The median values are nearly identical to the means (98.61 on reference and 97.425 on actual), confirming that the distribution of readings is consistent and devoid of significant anomalies.

The sensor consistently underestimates oxygen saturation by approximately 1.20%. The observed low standard deviation (approximately 0.34) reflects a high level of measurement stability. Our sensor achieves an overall accuracy of about 98.8%, slightly surpassing the 98.61% accuracy reported by Cebada-Fuentes et al. (2024) for an optimized pulse oximeter system. This indicates that the calibration and data processing protocols applied to our system provide reliable and repeatable measurements.

Table 4.

Accuracy of Temperature Sensor: Reference and Sensor Data Mean

Trial	Reference Data Measurement	Actual Data Measurement	Difference (%)
1	36.50	36.06	-1.21%
2	36.51	36.07	-1.21%
3	36.49	36.05	-1.21%
4	36.49	36.05	-1.21%
5	36.50	36.06	-1.21%
6	36.50	36.05	-1.23%
7	36.50	36.07	-1.18%
8	36.48	36.05	-1.18%
9	36.49	36.05	-1.21%
10	36.52	36.08	-1.20%
Total Mean:	36.49	36.06	-1.205%
Median:	36.5	36.05	
Standard Deviation:	0.011352924	0.011666667	

The temperature sensor's performance is detailed over 10 trials, with a reference mean of 36.5°C and a sensor mean of 36.06°C. The median values correspond closely with these means, confirming that the data is consistently distributed. The sensor shows a systematic underestimation of approximately 1.21% compared to the reference values.

The standard deviation, calculated at approximately 0.37, suggests minimal variability across trials. The accuracy of the temperature sensor is reported at 98.8%, which is slightly higher than the 98.44% accuracy observed by Santoso et al. (2023) using the MLX90614 sensor. This marginal improvement is likely due to rigorous calibration procedures and firmware optimizations that mitigate environmental interference.

Table 5.

Accuracy of QRS of ECG Sensor: Reference and Sensor Data Mean

Trial	Reference Data Measurement	Actual Data Measurement	Difference
1	90.25	87.82	-2.69%
2	89.93	87.51	-2.69%
3	90.32	88.15	-2.40%
4	90.76	87.79	-3.27%
5	89.88	87.01	-3.19%
6	89.88	87.44	-2.71%
7	90.79	88.15	-2.90%
8	90.38	88.24	-2.36
9	89.77	87.2	-2.86%
10	90.27	87.51	-3.05%

Total Mean:	90.22	87.68	-2.82%
Median:	90.26	87.65	
Standard Deviation:	0.361049	0.419227	

Upon testing the accuracy of CardioLink's QRS duration measurement, 10 trials indicate a reference mean of 90.22 ms compared with the actual sensor mean of 87.68 ms. The median values are close to the means, confirming that the data distribution is consistent.

CardioLink achieves an overall accuracy of 97.18%, confirming its ability to track the heart's electrical activity reliably. The low standard deviation for the proximity error of 0.304% reflects a high level of measurement stability and consistency across the 10 trials that were conducted. The stability is important because it ensures the minimal systematic error of 2.82%. The system ensures that, with calibration, it can meet the clinical standards that are being met. The overall 97.18% accuracy showed slight improvement over the 96.4% observed by Niu et al. (2020). Within those slight improvements, CardioLink's ECG showed a better performance from serious calibrations.

2. Statistical Comparison of CardioLink and Traditional Physiological Devices

Table 6.

P-Value and T-Values of CardioLink's Accuracy in the Reference and Actual Data Measurements

Source	Statistical Results		
	T-Value	P-Value	Decision
Heart Rate Sensor	2.26	$1.50 \cdot 10^{-18}$ or 0.00	Reject Ho - Significant
Oxygen Saturation (SpO_2) Level Sensor	2.26	$2.33534 \cdot 10^{-21}$ or 0.00	Reject Ho - Significant
Body Temperature Sensor	2.26	$1.62601 \cdot 10^{-18}$ or 0.00	Reject Ho - Significant
Electrocardiogram sensor (ECG) (QRS)	-19.6	$1.325 \cdot 10^{-8}$ Or 0.00	Reject Ho - Significant

A consistent t-value of 2.26 was found in the analysis for the body temperature, heart rate, and SpO_2 sensors. The corresponding p-values were 1.626001×10^{-18}, 2.33534×10^{-21}, and 1.50×10^{-18}. While the t-value of the ECG is -19.6. This value is larger in magnitude than the critical threshold. It statistically confirms that the 2.82% systematic bias in the ECG measurement is stable.

The null hypothesis is rejected for all sensors because these p-values, which are significantly below the traditional significance level ($\alpha = 0.05$), show that the observed differences between the sample measurements and the hypothesized values are statistically significant.

According to this finding, each sensor consistently identifies variations that are unlikely to be the product of chance. The surprisingly low p-values support the robustness of the findings, even if a t-value of 2.26 suggests a moderate effect size. Practically speaking, the findings strongly suggest that the effects (such as variations in body temperature, oxygen saturation, or heart rate) are real and not the result of chance fluctuations or measurement mistakes. The T-value of ECG, which is -19.17% proves that the 2.82% error in the device is real and highly consistent, which confirms it is the most stable and reliable sensor in the whole system. There are various implications for these findings. They verify that the sensors are identifying statistically significant differences, which could be helpful for research applications, clinical diagnostics, or health monitoring.

3. CardioLink's Response Time in Seconds

Table 7.

Response Time: Sensor Data to the on-board Display

Trial	Response Time (in Seconds)
1	2.02
2	2.00
3	1.93
4	1.89
5	1.97
6	2.00
7	2.05
8	2.06
9	2.06

10	2.00
Total Mean:	1.998
Median:	2
Standard Deviation:	0.05573748

The response time for the sensor data to appear on the device's 2.8-inch display was measured across 10 trials. Both the mean and the median response times are recorded at 2.00 seconds, indicating a consistent performance with minimal delay. The extremely low standard deviation of approximately 0.0557 seconds further confirms the uniformity of the display update time. This consistency in response time is critical for a real-time monitoring system, as it ensures that users are provided with timely and reliable information.

Response Time of the Sensor Data to the PHP-Based Database in Seconds

Table 8.

Response Time: Sensor Data to the PHP-Based Database

Trial	Actual Data Mean (in Seconds)
1	2.47
2	2.52
3	2.47
4	2.56
5	2.49
6	2.49
7	2.34
8	2.47
9	2.45

10	2.48
Total Mean:	2.47
Median:	2.475
Standard Deviation:	0.056411189

The transmission time for sensor data to be sent to and stored in a PHP-based database is evaluated over 10 trials. The mean transmission time is 2.47 seconds, with the median at 2.475 seconds, which demonstrates an almost identical central tendency. The standard deviation is very low (approximately 0.0564 seconds), indicating minimal variability in the transmission process. Such consistency in data transmission is essential for ensuring that healthcare providers receive current information promptly for effective remote monitoring.

4. Evaluation of the Criterion Participants

Evaluation of CardioLink's Sensor Accuracy

Table 9.

5-Point Likert Scale Evaluation of the CardioLink's Accuracy

Likert Scale Question	Mean	Standard Deviation	Classification
How accurate do you find the heart rate sensor?	4.06	0.57	Very Accurate (4)
How accurate do you find the SpO_2 sensor in measuring oxygen levels?	4.2	0.43	Very Accurate (4)
How accurate do you find the temperature sensor in measuring body temperature?	4.4	0.49	Very Accurate (4)
How accurate do you find the ECG readings, especially in the QRS	4.3	0.46	Very Accurate (4)

wavelength?			
Total Mean:	4.24	0.49	Very Accurate

The table above represents certain specialists' assessments of CardioLink's sensor accuracy based on a 5-point Likert scale survey in which participants rated the accuracy of the heart rate, SpO2, body temperature, and ECG readings in the QRS complex.

The heart rate sensor had a mean score of 4.06 and a standard deviation of 0.57, whereas the SpO2 sensor had a mean of 4.2 and a standard deviation of 0.43. The temperature sensor scored the highest, with a mean of 4.4 and a standard deviation of 0.49. The relatively low standard deviations indicated that CardioLink participants provided consistent responses, which added to the system's credibility and effectiveness. However, a similar study on popular wearable health monitoring devices found that, on a similar scale, user-rated accuracy scores ranged from 3.44 to 3.78 (Jia et al., 2018). According to this comparison, experts believe that CardioLink's sensors are substantially more accurate, which could be explained by its improved calibration and real-time processing capabilities.

Evaluation of CardioLink's Response Time

Table 10.

5-Point Likert Scale Evaluation of the CardioLink's Response Time

Likert Scale Question	Mean	Standard Deviation	Classification
How would you rate the speed at which sensor data is displayed on the device?	4.4	0.33	Very Fast (1-2 seconds delay)
How would you rate the speed at which sensor data is transmitted to healthcare professionals or family relatives?	4.06	0.25	Very Fast (1-2 seconds delay)
Total Mean:	4.23	0.29	Very Fast (1-2 seconds delay)

The table above shows the specialists' ratings of CardioLink's response time, based on a 5-point Likert Scale survey. The participants evaluated the sensor's response time as displayed on the device and transmitted it to healthcare professionals. The mean display speed was 4.4 with a standard deviation of 0.33, while the average transmission speed was 4.06 with a standard deviation of 0.25. Both were classified as Very Fast (1-2 second delay). The total mean score is 4.23 and the standard deviation is 0.29, indicating that CardioLink is efficient in terms of real-time health monitoring.

5. Implication of the Study

The implications of CardioLink in improving the speed and efficiency of measuring body temperature, oxygen saturation level, heart rate, and electrocardiogram readings are significant. With the automation provided by

CardioLink, the process of measuring physiological signs is greatly streamlined. Traditional manual methods for these assessments can be time-consuming, require multiple steps, and often involve human error, which can lead to inconsistencies in results. In contrast, CardioLink automates these processes, potentially reducing the time required for each measurement while maintaining high levels of accuracy and reliability.

In clinical settings, where time is often a critical factor, the ability to quickly assess multiple vital signs using a single automated system like CardioLink can improve workflow efficiency. This is particularly important in busy healthcare environments, where practitioners need to manage multiple patients simultaneously. The ability to save time on routine measurements allows healthcare professionals to focus more on patient care and decision-making rather than spending excessive time on manual data collection.

Moreover, the data captured by CardioLink can be instantly recorded and integrated into electronic health records (EHRs), further enhancing the efficiency of data management. The automation of these processes not only reduces the burden on healthcare staff but also reduces the potential for errors related to manual recording, such as incorrect data entry or loss of important information.

In research settings, the speed and accuracy of CardioLink also enable faster data collection across a larger sample size, enhancing the scalability of studies and potentially improving the overall quality of research. As more data is collected in a shorter period, researchers can analyze trends and patterns more effectively.

CardioLink's ability to automate physiological signs assessments has the potential to significantly enhance the speed and efficiency of routine healthcare operations. By minimizing the time spent on measurements and reducing human error, it not only optimizes clinical workflows but also contributes to better patient outcomes and research accuracy.

Chapter IV

Conclusion and Recommendations

Conclusion

This study evaluated the effectiveness and reliability of CardioLink, a real-time health monitoring system designed to track heart rate, oxygen saturation (SpO_2), body temperature, and the electrocardiogram of the heart (ECG). It aimed to determine the system's accuracy, response time, and overall usability as an alternative to traditional physiological monitoring devices.

To assess the accuracy of CardioLink, its recorded measurements were compared to those of standard medical devices. While minor variations were observed, the results demonstrated that the system provided consistent and precise readings suitable for continuous monitoring. In addition, the system's response time in displaying real-time health data was measured, focusing on both the onboard display and the transmission of sensor data to a PHP-based database. The findings indicated that although slight delays were present, CardioLink remained reliable in providing timely updates. Its efficient data transmission also allowed for the quick detection of abnormalities, ensuring that users and caregivers receive prompt alerts—making it applicable for both emergency situations and routine health tracking.

The system's accuracy and response time were further evaluated by professionals in healthcare, engineering, and information technology using a 5-point Likert scale. The standard deviation analysis revealed a consistent response, affirming that CardioLink is a highly accurate and dependable health monitoring system.

Moreover, its intuitive interface enhances accessibility, allowing medical personnel, caregivers, and patients to operate it seamlessly.

A comparative analysis with existing monitoring devices confirmed CardioLink's capability as a viable alternative. By integrating smart technology and wireless connectivity, the system ensures accessibility without compromising reliability. More importantly, it addresses gaps in healthcare accessibility by enabling proactive health monitoring outside hospital settings, reducing the strain on medical facilities, and promoting early intervention for at-risk individuals.

With its demonstrated accuracy, responsiveness, and ease of use, CardioLink proves to be a significant innovation in smart health monitoring. Further refinements can enhance its performance, ultimately improving real-time health tracking and accessibility for a broader range of users.

Recommendations

Based on the findings and discussions presented in this study, the following recommendations are proposed to enhance the effectiveness, scope, and reliability of the CardioLink system for real-time health monitoring:

1. **Expand Sensor Capabilities.** Such as those for blood pressure and respiratory rate, to provide a more comprehensive assessment of a patient's health. This expansion will improve the system's ability to monitor multiple vital signs, thereby enhancing its clinical utility, especially in settings where a full range of health parameters is necessary.

2. **Establish Routine Calibration and Maintenance Protocols.** Implement calibration procedures and scheduled maintenance routines to ensure that sensor accuracy is maintained over time. This will help minimize systematic errors, guarantee consistent performance under varying conditions, and sustain the reliability of the sensor readings.
3. **Enhance System Integration and Data Security.** Strengthening data encryption and privacy measures to ensure the protection of sensitive health information, thereby building trust among users and healthcare providers.
4. **Future Research.** Conduct future studies involving a larger and more diverse group of participants, including younger individuals and those with chronic conditions, to validate the system's performance across different populations.

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